GPW Evolution '98

What each change does, and why I made it

A plain-language tour of the mod, by Ferrim. Version 0.2, covering 0.1 and everything since

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What this is

I'll start with the TL;DR version: GPW Evolution '98 is a mini-mod for Grand Prix World that changes how the races, the AI and the paddock behave, and sets the game in the 1998 season. Everything in it is measured against Veedub's unofficial 1.03 patch, which is the version most of us have been playing for years. If something isn't mentioned here, it works exactly as it does in 1.03.

This is the consolidated narrative for version 0.2. It describes the whole mod against 1.03, including everything that changed after 0.1. Where something is new in 0.2, the heading says so or the paragraph starts with "New in 0.2"; everything else was already in 0.1, with the corrections that came later.

What you'll notice first, race by race:

- There is a Safety Car now, and the AI reacts to it - it pits, saves fuel, changes tyres, or decides it's better to stay out.
- The AI plays its race with a bit more sense: some cars start on the soft tyre, pit stop plans depend on the compound, fuel is loaded for the stint ahead and not for the whole afternoon, and a change in the weather makes them think instead of panic.
- The pit lane behaves. The old pit stop bug, where cars pitting at the same time would leave in 1998 car number order, is gone, and now the cars are visible on the map while they are in the pits.
- Chassis handling remains king, but no longer emperor. The parts, the driver and the set-up all matter more than they used to, and wet weather no longer decides a race on its own.
- The paddock moves. Contracts don't all expire at the end of the first year; they are randomized on game start. Free drivers lower their demands as the season runs out, and a driver who isn't delivering can lose his seat to a better one.
- The screens tell you more: a full race order with gaps, tyres in use; lap times of any driver; a more informative pitstop screen; qualifying margins against the 107% are visible now.
- New in 0.2: the AI handles rain and repairs much better. It re-plans when the track changes, splits its choices between team-mates, never pushes back a stop it had already planned, and a car that comes in for a repair uses the stop to change tyres and fuel.
- New in 0.2: penalties are applied where they used to fall through, parts wear where they should, the wind and the FIA regulations matter, and there are three new race screens: Live Championship Standings, Car Internals and a bigger Position tab.

The principle behind all of this is a boring, solid one, and I'm fine with that: keep whatever makes GPW feel like GPW, and fix what is plainly broken before adding anything new. The one real departure from that principle is the Safety Car. I got carried in a “no way this is going to work” moment, and it took some effort, but it ended up working well.

How to read this document

Each chapter covers one area and you can read them in any order. In every one I try to answer two questions: what does the change affect, and the reasoning behind it: why did I bother. Where it helps, I say how the original game did it, so you can tell what's new from what was already there.

There are no figures in the text: no percentages, no seconds, no lap counts, no money. That's on purpose. They're the numbers the game works with, so I don’t want to spill the beans. They have moved while I tested, and I'd rather describe a behaviour than freeze something that will be wrong next week.

Every feature in it has been through a test bench and through real races, and I'm happy with how it behaves, but this is a game with a lot of stuff, and you will find cases I haven't.

Many of the changes in 0.2 started from community contributions on the form, in the forum and on the Discord. Where that is the case I say so, with the name people chose to appear with.

What is in the package

The mini-mod is my EXE plus the files that go with it: the commentary lines, the audio, the pit lane maps, the tobacco sponsors and the like. You paste it over a clean installation of the game; you don't need the 1.01 or 1.03 patches. Start a new career to get everything: some changes, such as the contracts and the pay drivers, only apply to new games. DxWnd, which many of us use to run the game on a modern system, is not included, but the package brings a ready-made profile for it. Most of the graphics are Simul810's work and I have permission to use them. The new Race Control audio was generated with a text-to-speech model that is licensed for non-commercial use only, so take care.

The EXE is built from a generator that allows to switch on/off any single feature; I have plans to release an app which allows the players to generate their own exes at a later date.

The Safety Car

The original game has no Safety Car. A car can stop dead in the middle of the track, on fire for all the game cares, and the rest of the field goes round it at full speed as if nothing had happened. Yellow flags exist, but the way they are implemented in the game, they are there to hand out penalties, not to slow anybody down.

The SC wasn't as prominent in 1998 as it is now, but it was still used from time to time, and I thought it would be a nice challenge to implement one in the game. It is also the one that took the longest, because it touches everything: how fast the cars run, who is allowed to overtake, what the AI does in the pits, what the news says afterwards. So this chapter is a long one.

When it comes out

A Safety Car is called when a car stops on the track. The reason matters. A collision or a plain accident is the likeliest trigger, a driver error or a spin on his own comes next, and a mechanical failure is the least likely - an engine letting go at the end of a straight is not the same kind of problem as a car broadside across the circuit, and so the game treats them differently. A car that stalls on the grid can also bring it out, in any circuit, and it is the only trigger that works before the race has really started. A car retiring in the pitlane will never trigger a SC.

Each stopped car gets one roll, once, and that's it. If it doesn't bring out the SC, nothing will make it do so later. That keeps a long race from turning into a parade just because a few cars retired one after another.

There are limits on when it can appear. It can come out from the very first lap, because first-lap incidents are exactly where it belongs, but not when the leader is close to the finish - there is no point in neutralising the last moments of a race, and I'd rather see the order settle on the track. And it won't be called twice at the same time: once it is out, further incidents are simply noted and wait their turn.

How long does it stay out? A few laps of the leader, with a bit of randomness so you can't set your watch by it, and always trimmed so that it ends comfortably before the flag.

What the race looks like under it

The moment the SC is deployed, everything slows down, and this is where most of the work went.

The leader drops to a crawl. The rest of the field slows too, but not in the same way: a car that is far from the one in front is allowed to close up, and a car that is already close just holds its distance. The result is that the pack bunches up behind the leader without anybody running into anybody, which is harder than it sounds. Slow everybody down by the same amount and the gaps never close; let the whole field catch up freely and the cars pile onto the same stretch of track, and the race order turns into a lottery. The in-between is what makes it look right.

Overtaking is off for everyone. The only exception is a car that is stopped or retired, which you can of course go past. When a car is told it can't overtake, the game makes it copy the pace of the one in front, and that is what forms the queue.

A ghost car stands in for the real Safety Car. It leaves the pit lane almost standing still until the leader catches it up, which gives the whole thing a natural start instead of an abrupt wall of brake lights. On the last lap of the Safety Car period the leader slows down a bit more and then, at a point picked at random, goes again. It's a small touch, but it means the restart isn't the same every time.

On the circuit map you'll see a silver circle with "SC" in it, right where the ghost car is. And Speech Transcription tells you when it has been deployed, when it comes in at the end of this lap and when the green flag is out. If you don't have the commentary files installed, the SC still works - it just does so in silence, without informing the player.

While the SC is out, the cars burn much less fuel, and their tyres and parts wear far more slowly.

The laps spent under the Safety Car also count when the game decides when to serve a penalty. A penalty that was supposed to come due while the field was crawling doesn't get served until after the SC is already gone.

The ghost car and the queue, tuned (new in 0.2)

The first version worked, but there were some bugs. The ghost SC sometimes seemed to jump backwards just before it came in, and in some races a few cars were left far behind when the queue should already have formed. Two things were wrong: the ghost kept its own pace while the leader braked on the last lap, so it ran away from him until the game decided it was too far and put it back; and the cars that were far from the one in front weren't closing up fast enough.

Now the SC brakes together with the leader on the last lap (just a bit less), the limit before the game moves it back is wider, and cars that are far behind close up quicker. The distances between the cars while the SC is out, and the gap at the restart, are a bit smaller than in 0.1: with the old ones, the field spread was a bit too big for my liking. I’m still not 100% happy with the results, but it’s a work in progress.

Blue flags (new in 0.2)

A lapped driver sitting between two leaders behind the SC was being investigated for ignoring the blue flags. Under the SC nobody can overtake, so nobody can be blamed for not letting anyone by. That check is switched off while the SC is out. Blue flags messages may still pop up, and when there are lapped cars behind the SC, I’ve found that the first few laps there might be a bit too many penalties, so maybe in a later release I’ll decrease the penalty odds for the first couple of laps after the SC.

What the AI does about it

When the Safety Car comes out, the AI stops following its plan blindly and works out what is best for each car.

If a car is running low enough on fuel, it refuels at once. If its planned stop was coming up anyway, it goes in without looking at the gap to the cars behind - when the stop is close, the Safety Car is the best moment to take it. If it doesn't have a stop coming up, it compares what it would lose in the lane with the distance to the car behind: if that car is far enough back, the stop is effectively free and the car goes in, otherwise it stays out.

A car can end up stopping once less than planned because the Safety Car lent it the fuel it needed. Another can come in early and finish the race on a stint it was never going to complete. In the news after the race you may see a few of these stories told: a team that pitted because it was about to run dry, a team that brought its last stop forward and filled to the end, a team that saved a stop thanks to the fuel the Safety Car had spared. They're drawn from what actually happened in that race, with a few different ways of telling each one so you don't read the same sentence every time.

Two details I should mention

Cars in the pit lane keep their own speed limit under the Safety Car. I didn't want the lane to slow down to the same crawl as the track, because a car in the lane isn't part of the queue. The lane has its own treatment and I describe it in the pit lane chapter.

The times used for a stop under the Safety Car are, for now, provisional. I know roughly how long the stop and the lane cost, but I haven't measured them circuit by circuit, so the AI's idea of "is this stop free?" is a good estimate and not an exact one. If you see a car make a decision that looks a bit off, that's probably the reason, and it is on my list.

Why I did it this way

The lazy version would be to slow everyone down, switch off overtaking and call it a day. That isn't a Safety Car, more like a Virtual Safety Car, which didn’t come in until much later in real life. What makes it work is that every part of the race has a reaction - the pace, the queue, the pit decisions, the penalties, the map, the news. If you remove any one of them, the rest feels fake.

I'll say it plainly: the Safety Car is the feature in this mod that I'd most like you to try and break. It has more moving parts than anything else, and the more circuits, weather conditions and strange race situations it sees, the better I'll understand where it goes wrong.

How the AI plays its race

If you've played GPW for long enough, you know the AI's habits by heart. Every car starts on the same tyre. The number of 1/2/3 stops is the same for every circuit, and the laps are more or less at about the same laps. When it starts to rain they all run for the pits, although some will stubbornly stay out on dries (and the other way around).

This chapter is about the changes made to those and a few other, not so well-known behaviours, that give the AI a bit more of a head for racing, and most of them are small ideas with a long reach.

Some cars start on the soft tyre

In the original game, every AI car started on the hard compound. Always. The soft tyre existed, but only the player ever used it, so a good part of the strategic game was missing for the rest of the field.

Now some of the AI cars start on the soft. The draw is made for each car on its own, once per weekend, and it is remembered, so a car doesn't qualify on one compound and race on another. When a new weekend begins the draw is cleared and made again. The soft tyre also loads its own attributes now: before, the AI's soft tyre was in practice a hard tyre with a different name, because the grip and temperature values that make it a soft were never loaded for the AI. Of course, it didn't matter because of the aforementioned the-AI-never-used-the-soft thing, but still needed to be fixed.

I wanted the proportion to feel like a choice and not like a rule: you will see a few soft starters, never the whole grid, and never the same cars. Right now, the game doesn't inform you about who is on which tyre compound, but that might change in the future.

A plan that depends on the tyre

Because the AI always took on hard tyres, the game used to decide how many stops an AI car would make without looking at what it was running on. Now the plan follows the compound. A car on the soft makes more stops than one on the hard and never gets away with a single stop, while a car on the hard can go far longer. Pretty obvious stuff, but the original game didn't do it.

The length of each stint follows the same logic. A soft-tyre stint is capped to a part of the race distance, so the tyre isn't asked to go further than it can, while the hard-tyre cars are allowed to stay out a good deal longer before their first stop.

Monaco gets its own treatment. It's a circuit where passing is close to impossible and every stop costs a lot, so I didn't want to see the AI follow the same pattern as everywhere else. At Monaco, the AI will try to stop as few times as possible, with the predictable consequences in terms of tyre choice and number of pitstops. It's likely I'll extend this pattern to other tracks in the future.

Fuel, at the stop and at the start

There were two problems here. First of all, at a stop, the AI added a whole stint of fuel on top of whatever was left in the tank. So a car that came in with a lot in hand would leave with a ridiculous

amount, and the extra weight made it slower than it should have been. Now it only adds what it's missing to reach its plan, and no more than what it needs to finish the race. The laps it still has to run are counted from the leader's lap count, so a lapped car works from the same basis as the car in front of it and doesn't put in extra fuel it won't need.

And compounding that, every AI car started the race with the same number of laps worth of fuel. A car planning an early stop carried as much as one planning to go long; the game "saves" the AI from running out of fuel, and this is what was behind those weirdly quick 1-stoppers that we have all suffered while playing Grand Prix World. Now, each car starts with the fuel for its own first stint. It's a small change, but one that makes the race look much realistic and accurate.

The weather runs on the clock (new in 0.2)

In 0.1, qualifying and the race handled the weather in different ways: in qualifying it depended on the session time, and in the race on the leader's lap. That meant the track always changed at the same place, when the leader crossed the line, and you could wait for it. Now the race weather runs on time, as qualifying does. It will still change, but not at a moment you can predict.

When the weather changes

When it starts or stops raining, the AI re-plans its stops. That was always the case, but the re-plan had problems, and getting it right took more rounds than anything else in the mod.

The first stop after the re-plan used to be set too late, and sometimes the car skipped it altogether, meaning it would stay out on the wrong tyre for the conditions. Now it comes soon after the re-plan, so a car on the wrong tyre doesn't carry on for several laps. A similar thing happened when an AI found his teammate in the pits and had to stay out: it would never pit to take on the correct tyres. Teammates that can't pit because the pit lane is occupied will pit the next lap. The thresholds that decide how many stops a car makes after the re-plan were fixed lap numbers, which only really made sense for a race of one particular length; now they scale to the real distance of the race, and the laps remaining are counted from the leader's laps.

New in 0.2: a few more things came out of watching real races, and several of them were reported by you.

- The in-between case. The AI used to ignore the moment when the track would change from wet enough for inters to wet enough for wets, and the other way around. Everybody stayed out, whatever tyre they were on, although they would pick the correct tyre at the time of their scheduled pitstops. The player could exploit this, although from time to time, the weather would change quickly enough that any advantage obtained would be erased, and occasionally it would even back fire.
  Now, each AI team splits its choices: the first driver gets the choice to pick the tyre that suits the new state of the track, and his team-mate gets the same chance if the first one refuses. This idea came from the community (Acescheil).
- A re-plan never pushes back a stop the car had already planned. This came from a car that had a stop coming up and ended up stopping later because the weather changed, as if it could magically carry more fuel.

- The ideal tyre is the same for every driver, whatever his wet-weather skill: intermediates for damp, wets for wet. A driver who complains that he wants a different one doesn't change what is best for the car.

No stint too short

One more thing that bothered me is that the plan for each AI car was drawn stop by stop, and nothing looked at how long the stints between the stops came out. In a good number of races and under particular weather change scenarios, that meant stints too short that no one would plan in a real race.

Now there is a floor. No stint of an AI car can be shorter than a fixed share of the race distance, rounded up: not the first one, from the start to the first stop; not the ones between two stops; and not the last one, from the final stop to the flag. It is applied to the plan the car is given at the start, after the rules for each tyre, and it is also applied whenever a re-plan happens.

A stop that was getting lost

As already explained for the case of weather changing, when a car was due to stop and its team-mate was already in the pit lane, the game was supposed to make it wait one lap, but it compared the numbers wrongly and the stop simply vanished. Worse, the car would also skip a stretch of the track, making up some time and setting unrealistic lap times in the process. Both behaviours are fixed now.

The broken front wing

An AI car with a broken front wing used to keep going round with it, as if it hadn't noticed. Now, in the race, if the wing is badly damaged and there are enough laps to go, the car comes in to have it repaired. It won't keep trying for ever: a small number of attempts, and that's it. You'll see this more than you think, mostly in the first laps.

New in 0.2: in 0.1 the car did come in, but the pit crew never got the order to fix anything: it stood still, left, and often retired soon afterwards. Now the wing is repaired for real. The same goes for any other breakdown, with a twist that came from the community (Acescheil): a car that is running in the points takes the risk and stays out, while one that is running down the field comes in to have it checked. The repair itself can still fail. That is how the original game was built, and I haven't touched it.

A stop for a repair is a stop (new in 0.2)

A repair stop is also a stop: the car is in the pits anyway, so it used to be a waste not to change its tyres or take on fuel. Now the AI treats the unscheduled stop as the first stop of a new plan. It refuels, changes tyres, and plans the rest of its stops again from the laps that are left. Whether it makes sense for it to finish with one stop fewer than it had planned depends on the race, the same way the weather re-plan decides.

Qualifying: when the cars go out

The original game had a rotating pointer and a coin toss to decide who left the pits in qualifying. There was a limit on how many were out there at once. It worked, in the sense that it produced a classification, but oftentimes didn’t look like a real session: AI cars would waste laps on a soaked track, and a car outside the 107% because the session started under the rain would carelessly stay in the pits until the rain came back.

Now, each run is scheduled according to how good the car is - slow cars go out early, fast ones wait for a better track. There is no limit on the number of cars on track, so the pit exit looks like a pit exit. And the AI is sensible about the weather: if the track gets wetter than it was when the best time was set, the cars stay in, unless it's the final stretch of the session and they have nothing to lose. When the target time was set in the rain and the track is still dry, the cars without a time go out first, on a rush, which is what you'd do.

New in 0.2: when it started to rain in a session, the whole AI used to leave the pits at the same time, and a driver coming back after a bad lap could end up waiting in the pits and not run again. The AI leaves at spread-out intervals now, and a car that is outside the 107% can go out again. The state of the track at the moment the pole is set is also recorded at that moment, because there was an edge case (when the weather changed at around the time the pole was set) that the AI thought the pole time was set on a drier condition than it actually was, meaning it would keep sending its cars out pointlessly.

A small fix with a long reach

This last one isn't about strategy, but it belongs here because it affects how everyone behaves. The constructors' table sorts teams properly - by points and then by wins, seconds, thirds and so on. What it got wrong was the position it stored for teams without points: they all received the same number. That stored position feeds the share prices and the news, so after the first race with only a few teams scoring, all the others were "the same position", and the prices came out inflated. Now every team on no points gets its own consecutive position, following the table's order. The screen still draws the "=" for them, and the teams that have points behave just like they ever did.

The pit lane

The pit lane in the original game has a reputation, and not a good one. You've probably seen the effect without knowing what caused it: a car that entered the lane first leaves after one that came in later, when the later one has a smaller number. That is the pit stop bug, and it's the first thing this chapter deals with.

The pit stop bug

The cause is that the cars stopped in the pitlane were ordered by a common key, with the car number as the tie-break, and each car waits behind the one in front of it in that list. If two cars were stopped at the same time, the one with the higher number would be made to wait for the one with the lower number, even if it came in first. Hence the odd departures. When it happened in a race, it also gave the player a very unfair feeling: you do everything right and somebody jumps you in the pits.

In this mod the cars that are in the pit lane together no longer wait for each other, either on the way in or on the way out. The bug is gone now.

New in 0.2: this fix was in my early test builds and dropped out of the 0.1 package by mistake, so 0.1 went out without it. It is back. If you still see a car with a low number jumping one with a high number in the lane, I'd like to hear about it, with the circuit and the lap.

A speed limit that means something

As far as I can tell, the lane in the original game has no real effect on the pace of a car: it runs through at the speed of the track, which makes a stop a lot cheaper than it should be.

Now cars slow down in the pit lane. The slowdown is a factor applied to the pace of the car while it is in the pitlane sections, and it's different for each circuit, because the lanes are different: a short, direct lane doesn't cost the same as a long and winding one. The time a car spends in the lane is the time it takes to go through, which depends on that factor, plus the time it is stationary.

I calibrated it against real pit stops from the late nineties, circuit by circuit, with the times measured inside the game. For the circuits that have been calibrated, a stop costs about what it did in real life. The ones I haven't measured yet use a general value that's a sensible compromise, and I'll tighten them as I get to them. A quick note on the way I measured: all my timings use the game's own internal clock.

New in 0.2: after the first races with 0.1, I compared the stops with the real ones again. In some circuits a stop was costing too much and in others too little, so Australia, Brazil, Spain and Austria now have values of their own, adjusted in one direction or the other. The rest keep the general value until I get to them.

The Safety Car doesn't change this. The lane keeps its own treatment under yellow, so a car in the lane isn't slowed by the same crawl as the cars on the track (making it beneficial to take a chance to pit).

Cars in the pit lane, on the map

The race map used to show the cars going round the track, and the moment one of them entered the pit lane it disappeared. It was actually still there, but the map had no idea where the lane was.

The new maps do. Each circuit has its pit lane drawn with its sections and its points, and the cars are visible all the way through the stop. The maps also carry a marker for where the Safety Car parks, so the ghost car has somewhere sensible to go. Old maps still work: if the game finds one without the new information, it behaves exactly as before.

The Safety Car itself has a spot on the map while it waits. At first it appeared almost at the point where the lane rejoins the track, which looked as if it were already leaving. I moved it back along the lane, on every circuit, so it sits where it would on the real track (more or less).

What makes a car fast

Every racing game has a recipe for pace: some amount of car, some amount of driver, a pinch of set-up, and a few things that happen on the day. GPW's recipe is a strange one, as the community found out many years ago: the chassis and the engine power together explained nearly everything, the driver was a rounding error, and the rain was a bigger factor than the whole car. That isn't how racing works, and it isn't how it feels either, and finally, it made for a poorer game; so a good part of this mod is about putting the ingredients back in proportion.

The parts matter

A GPW car has seven developable parts: gearbox, clutch, suspension, throttle, hydraulics, electronics and brakes. In the original, they all counted in the pace, but so little that you could spend a whole season developing them and barely notice. The chassis curve swallowed everything else.

In this mod every part weighs a lot more. The same goes for the four driving aids - assisted brakes, traction control, the automatic gearbox and the active suspension - and for the fuel and tyre supplier. The relative importance of each piece stays the same as in the original: the suspension still counts for more than the hydraulics, because the game said so and I didn't want to second-guess it. What changes is the overall volume. A car with a good set of parts is clearly faster than one with a poor set, which is exactly what you should expect from a development programme.

The engine curve and the starting engine ratings are described in the chapter on the 1998 data.

The driver matters

In the original, a driver's speed and skill together moved the pace by a tiny amount compared to the chassis. The difference between the best driver on the grid and the worst was, for practical purposes, noise.

Now both count much more, and they count relative to a middling driver. A driver with an average rating is exactly where he was before, and the further he gets from that, in either direction, the more his pace changes. It's a gentle slope, not a cliff.

There's one more piece, which is the one I'm most pleased with: a good driver helps a bad car more than he helps a good car. In a car that is already strong, there isn't much left to gain, while in a weaker one a talented driver can hide a fair amount of the car's faults. This is why you'll see the occasional driver punching above the team's weight. It also, for whatever reason, worked the other way around in the stock game.

Pushing

Push is the order to go flat out for a lap. In the original it gave a modest pace bonus, and its cost was tiny, so the AI and the player alike found it a bargain.

Now the bonus is bigger, and the bill is bigger too. A push lap raises the chance of a mistake, wears the engine, wears the tyres more and heats them up more. The engine wear is the one to watch:

every push lap now costs the engine some of its life, every single time, and it can't get any worse than that. So push is a tool you use for a lap or two when it matters - an out-lap after the stop, a gap to close - and not a permanent state. Which, to be fair, is how it should have been from the start.

Tyres and set-up

The stiffness of the tyre counts for a lot more in the lap time than before. It was already there, but its weight was low enough that nobody could tell. The wear and the grip of the tyre keep the original weight, because the original was fine and I'd rather leave a thing alone than "improve" it for the sake of it.

New in 0.2: there is one exception to "leave it alone". In the original, the grip of the tyre made the car better up to the middle of its scale and then did nothing, while still costing fuel and tyre temperature. That looked like a mistake and not a design, so the grip keeps giving something now.

The set-up does what it always should have done: it makes a difference. The points you spend on the speed and grip of the set-up weigh more, and the tables behind them are steeper, so a car with a good set-up is visibly better than one without. As with the drivers, they're measured from a middle value, so a car that starts at the circuit's default set-up isn't pushed out of its place.

New in 0.2: the set-up reserve is capped. You could end up with more points than the game was meant to hold by testing, applying and taking them back.

Wet weather

This is one of the changes I wanted the most. In the original, the weather on the day was a force of nature. A driver's wet-weather rating, together with the rain set up of the car, could swing the lap time by an amount comparable to the whole spread of chassis performance. It meant that, in the rain, a car's real quality hardly mattered: what counted was who had the right stars on the right day.

Now both factors are much weaker, a fraction of what they were. They are still there, and a driver who is good in the wet will still do better than one who isn't, but it doesn't take over the race. The reference point stays where it was: a middling wet-weather driver with the car's default rain set-up loses exactly what he lost before. Only the distance between the good and the bad has shrunk.

I don't think rain should be a neutral element, and it still isn't. It is that it shouldn't be almost the only thing that matters when it falls. If it rains and the best car wins anyway, I consider it a good sign. If it rains and an outsider surprises everybody, that's racing, and I'm happy with that too, as long as the surprise was earned. If I find out the effect is too small now (I went pretty much out of my way), I will increase it again.

One circuit out of step

There is a value for each circuit that sets how far apart a good car and a bad car are, in speed, when they race on it. Fifteen circuits had values that were close to each other. Germany did not: its value was far from the rest, and the effect was a field squeezed together, where the car at the back was almost as quick as the car at the front, and the weight of fuel hardly seemed to matter.

I have brought Germany in line with the others. The spread you see there now is the same kind of spread you see everywhere else, and it agrees with the one the game's own quick simulation always used. I won't pretend the lap times are now perfect: with the spread corrected, the overall pace of the circuit is a separate question, and I'm still looking at it.

Overtaking (new in 0.2)

The speed bonus in an overtake used to depend on a ratio that nobody understood. Now the faster car gets a bonus that depends on how much faster it really is than the other one, with a limit. The Overtaking value of the circuit counts only for the attacker: before, it was added to both attacker and defender, meaning the track rating made no difference. How much that value matters depends on the circuit, and I'm still tuning a few of them.

The wind (new in 0.2)

The wind existed in the game, but what it did was close to the opposite of what you would expect: the more wind you set up for, the more you lost, and with some settings you lost nothing at all. Now the penalty depends on how far the wind setting of the car is below what the real wind asks for. A car with enough wind setting loses nothing; a car with none loses a clearly noticeable amount when it blows hard. This is a first version, and I'll take another look at it once more of you have played with it.

The FIA (new in 0.2)

The FIA regulations come up every year in the game, but they almost never changed anything. Three things have changed:

- The caps for engines, fuel and tyres are lower, so the regulations actually act on the suppliers.
- When a chassis regulation is approved with continuity, the design now starts by itself with the progress that came with it. Before, pressing the button lost it.
- The type of the regulations moves the base pace of the circuits: when nothing changes the cars get faster as the teams keep developing, and with a change they get slower.

Wear, failures and the two-hour limit

This chapter is about how things get tired, break and end, which in GPW is a bigger deal than it sounds: the whole rhythm of a race is a negotiation between going fast and arriving. Most of what follows exists because the original numbers were calibrated for one kind of race and kind of fell apart in the others.

Wear follows the distance

In the original, tyres and parts wear by the same amount on every lap, whatever the length of the race. A short race and a long one therefore wear a car by completely different amounts, and the shorter circuits were far too kind on everything. Now, for the case of chassis wear, the game included a scaling factor which partially compensated for this, but it didn’t go far enough.

Now the wear per lap is scaled to the distance of the race, so that a race wears a set of tyres about as much as the reference race does, whether it is long or short.

I should add that parts already had a character per circuit in the original - a circuit that is hard on brakes was hard on brakes - and that stays. What didn't exist was the dependence on the length of the race, and that is what has been added.

The table of rates, redone (new in 0.2)

Brazil was always hard on cars. After 0.1, Germany and Belgium turned out to be just as hard, to the point that a small team could not reach the end. The cause was the way I had scaled wear by the distance of the race: on those circuits the scaling stacked on top of an already high rate. The table of rates has been redone with the scaling built in, and the closing of the session uses its own rates now too. Germany and Belgium go back to something a small team can survive, Brazil is a bit kinder, and the circuits that used to wear very little wear a bit more.

Failures that behave, with a safety net

Random failures are at the heart of the original, and they remain. What I added is a gentle correction. At a few set points in the race the game looks at how many cars have retired for a mechanical reason, compared with the cars still running, and if the field has been unusually healthy, it may pick a healthy car and give it a failure, drawn with the game's own probabilities for what breaks. Each of those checkpoints is evaluated once per race, and it doesn't turn into a massacre.

Why? Because a race with no failures at all is the kind of thing that happens in a simulation and almost never on a real 1998 afternoon, and the other way round: if too many cars have dropped out already, the game leaves it alone. The idea is that the number of retirements should be believable from race to race, without fixing the outcome of any one of them.

The original also had a cap on retirements per circuit, a table of limits that stopped the race from losing too many cars. I've taken the cap away. If a race is a bloodbath, so be it.

A fairness fix goes with this. In the opening laps of a race the chance of a failure is raised for everybody, but the original exempted the human player from that. Now you get it too. It's the same game for everyone.

Monaco (new in 0.2)

I had asked myself why so many races at Monaco ended with almost everybody still running. Parts could only fail when a counter reached zero, and some rarely did. Monaco now wears the parts of the cars more, and a healthy car has a small chance of hitting the barriers at the end of each lap. That includes your own cars. The SC treats it as an accident.

The AI and its driving orders (new in 0.2)

In the original, the AI only started easing off its driving orders when a part was almost gone. Now it does it in steps: as the counter of a part falls, the target order falls with it, one step at a time and never more than one step per lap, for brakes, suspension, clutch, gearbox, hydraulics and throttle.

Parts and reliability (new in 0.2)

The starting counter for reliability was the same for the two highest ratings in every part. That isn't the case any more, so you'll need to be more careful with your orders when your reliability is lower. The design department had some similar slips: the progress counter of a technology part wasn't reset when the part was finished or at the end of the season, so the next one came out already complete; the CAD building did nothing; and the four stages of the chassis design weighed in the opposite order to how long they take. All of that has been fixed.

The two-hour limit

GPW has the real F1 rule that a race can't go on for more than two hours. But the original enforced it in a strange, buggy way: as soon as the leader's own clock reached the limit, that car was declared finished, wherever it was. It never had to cross the line, and the classification came out in the physical order on the track - which meant the car in second place could be a lapped one.

Now the clocks keep running and the race ends the way a real one does: when the leader crosses the line after the time is up. Everybody else finishes the lap they are on. The commentary has new lines for this, and there are new audio files to go with them, so you hear the announcement as well as read it (although the sound quality isn't the best).

New in 0.2: a car that has already crossed the line after the flag no longer loses its result if something breaks before the race is officially over. In 0.1, as in the base game, a car that broke down or crashed after taking the flag, while the others were still finishing their lap, was counted as retired, and its place and its points went to the car behind. Until the race is over, the panel may still show it as retired.

Pit stop times

There were two problems with the stop times. The first: the game raised every stop time to a minimum before applying it. Because of that, a plain tyre change took longer than it should have, and the stops came out all over the place depending on the circuit. The tyre table, which says how long a change of tyres takes, was effectively ignored. The minimum has been brought down, so now it only matters for a stop with no refuelling, no tyres and no repairs, and the table does its job again.

The second: the statistic of the best and the worst stop of the race rounded the fast ones up to a minimum, so it never showed anything faster than that. It now records the real time. This one only changes what you see on the screen, not what happens on the track, but I got tired of looking at numbers that weren't true.

Penalties

Penalties in the original game are immediate and a bit absurd. A car does something wrong - a false start, speeding in the pit lane, ignoring a blue or a yellow flag - and the penalty appears on the spot, as if the stewards had been standing next to the car with the paperwork already filled in. There is no investigation, no waiting, no sense of a decision being taken. This chapter is about giving penalties a process; for the time being, it is just for flavour, but I intend to make it probabilistic at some point.

An investigation, then a decision

When a car earns a penalty during the race, it isn't served at once. It is held back for a few laps while it is "investigated", and then applied. The delay is a little random, so you can't count the laps. During the wait, the game lets you know that an investigation has been opened.

This matters in practice for two reasons. First, the drama: you feel the tension of a car that might be punished, and you start thinking about whether the driver in front will get away with it. Second, the consequences: while the investigation is open, the car carries on with its race, and it only pays when the decision comes.

When the penalty is finally applied, it becomes either a stop-go that the car has to serve, or a time added to the car's result. If the decision arrives very late in the race, when there's no room left to serve a stop-go, it turns into a time penalty added at the end. You won't see a driver cross the line first and then be told he owed a stop-go that was never possible.

Serving it

The AI doesn't dive into the pits the moment the penalty is applied. It comes in a few laps later, playing with the three lap limit as a real team would. The human player will see his driver pit in automatically on the third lap after the penalty is applied; it’s up to you to bring forward its fuel stop if you see that it risks running out of fuel (because penalties and pitstops cannot be served at the same time).

The classification takes the time penalty into account. When two cars finish on the same lap, the order is decided by race time plus the penalty seconds, and not by race time alone. The time shown on the screen doesn't change - I didn't want to alter what you read - but the order is the right one.

New in 0.2: penalties are now applied in a few places where they used to fall through.

- A stop-go that had been given but not served by the end of the race now counts in the final order.
- If a penalty is enough to drop a car behind cars with fewer laps that finished behind him, he now drops behind them and counts as lapped. The Race Order panel shows it: how many laps down, and the penalty in red after the name.
- If you have a stop-go to serve, you can now call the pits freely during the first laps after it, to refuel and change tyres. When the time to serve it comes, the car comes in by itself, as before. Before, "Pit now" was blocked for the whole period.

Speeding in the pit lane

The original game ignored speeding in the pit lane in the closing stages of a race. I think that's backwards: the end of the race is exactly where a mistake decides something. Now the limit is enforced for the whole race.

Fewer reminders

The manager's spoken penalty announcements during the race are switched off. They belonged to the old instant penalties, and the investigation notice does the job now.

Illegal driving aids

The game lets a team run its driving aids beyond what has been approved, and it checks for it now and then: a driver is picked, found out, excluded from the session, and the team loses something. I like that, and I wanted it to stay. The problem was how often it happened to the AI. As the AI teams developed their aids season after season, they were caught again and again, drivers kept disappearing from the classification after qualifying, and it spoiled the fight for the title.

New in 0.2: the AI is also much less reckless about fitting illegal driving aids in the first place, and the risk it takes falls as the season goes on.

Now an AI team is found out a good deal less often. Your own team is checked exactly as before, so if you want to take the risk, the risk is the one the game always had.

How everything fits with the Safety Car

If the Safety Car comes out, the laps spent behind it count towards the delay before the penalty is applied. A penalty that was due while the field was crawling around is pushed back, so nobody gets punished at a moment when it makes no sense to serve it. The time penalties don't move, because their deadline is already tied to the closing laps.

New in 0.2: there are no investigations for ignoring blue flags while the SC is out. Nobody can overtake behind it, so nobody can be blamed for not letting anyone by.

The market and contracts

The paddock in the original game has a dull, mechanical quality. Everybody begins the game with a one-year deal, so everything expires on the same day at the end of the first season, the AI signs drivers more or less without looking at its own bank balance, and a driver left without a seat will ask for more money every time you look at him. Nothing happens in the middle of a season. It all feels like a large machine clicking once a year.

I wanted the paddock to move at a more natural pace, with people arriving and leaving at different times, teams that can only afford what they can afford, and a driver who isn't delivering being under some pressure. None of it is dramatic. It's the sum of lots of small decisions that mostly come out right.

Contracts that don't all end together

When you start a new game, contracts are no longer all for a single year. They are shared out by group, and each group has its own pattern.

Pay drivers - the ones who bring money to the team - always start with a single year, because that's the nature of the deal. Race drivers are spread over short, medium and long contracts, with the long ones the rarest. Test drivers lean towards the short end. Department chiefs, sponsors and suppliers lean short as well, with a few longer deals mixed in.

The effect is that the first winter isn't a blood bath any more. Some people are free, some are not, and the next year something different happens. It also makes the long game more interesting, because a contract is now something you plan around.

For the future, I might add a “real world” option where contracts mirror what happened in real life, at least through the end of 2000 (three seasons).

Who the AI can afford

In the original, the AI had flags about money, but they were rather permissive, and a team with a modest income could end up with a very expensive driver. It also didn't "look for" pay drivers: a big team only ended up with one if every better candidate had turned it down.

Now the AI signs a driver only if his pay is no more than a share of what the team earns. The share grows with the income of the team, which means a rich team can spend more, but it isn't a straight line: the share is small for a modest team and is capped for the richest ones. A team that is short of money can still take a pay driver, and the richest teams are kept away from them, which is what you'd expect.

The practical result is that the biggest salaries end up where there is money to pay them, and the teams at the back make do with cheaper or paying drivers. It also stops the strange cases where Minardi signs Hill, and the like :)

Free drivers who lower their demands

A driver without a seat used to ask for his previous wage plus some more. Free agents never became cheaper, so a good but expensive driver that nobody could afford simply stayed free for ever.

Now a driver without a seat for the following year starts from a discounted wage, and in the last rounds of the season he lowers his demands, a bit each time, together with his bonuses. This doesn't touch the contracts that are already signed, it doesn't affect pay drivers, and it doesn't apply to the player's team - what you offer is your own business.

The close of the season

At the end of the season the AI used to do a single round of signings. Now there are several, and they get less strict as they go: the first one is the normal one; the second one leaves out the special treatment of pay drivers; and the last one has no budget cap at all. The aim is simple: not to leave good drivers without a seat for silly reasons, while still giving priority to the sensible deals.

One last detail, for the pedants: when two candidates are exactly level for a team, the cheaper one goes first. A pay driver, whose wage counts as negative, goes before a driver who has to be paid.

Mid-season changes

Between the early and the late part of the season, after each race, a free star - a driver with serious talent and no seat - can sign for the best team that has a driver performing well below what his team's position deserves. The weak driver is released, with a news item that names the three people involved, and the star signs for a discount on what he wanted. It happens at most once per race, and the AI doesn't pay a penalty for the termination.

The replaced driver isn't picked at random: he has to be clearly underperforming, and the replacement has to be a clear step up. It adds a texture the game didn't have, and it gives a reason to care about a team's second driver.

I plan to extend on this feature: cases such as Stewart letting Magnussen go, as was the case in 1998, should happen once or twice per season, but that’s not currently the case.

After qualifying: who the FIA lets in

The original game lets the FIA readmit drivers who had been left out of the qualifying cut. It had three rules to do so: drivers who had suffered a problem, if they belonged to a team near the top; the test drivers; and a random quota of pardons.

Two of those rules are gone. The one for the front-runners and the one for the test drivers always felt like favouritism, and the cars they protected were never the ones that needed it. What is left is the random quota, which no longer asks for any particular condition, and the car in last place is always kept out. If you can't qualify, you can't qualify.

Smaller changes in the paddock (new in 0.2)

- **Salaries:** the cap on what a team can pay, based on its income, now also applies to department chiefs, and there is a second pass at the end of the season.
- **Test drivers:** the fixed veto of the original game, which kept three drivers from ever being signed as test drivers, is gone.
- **Pay drivers:** because of the bug with GPWEdit 1.1, the game was paying them instead of being paid by them. That is fixed for new games; a saved game keeps its own data.
- **Repairs** use the skill of the mechanical staff, not the commercial staff. That came from Jose21crisis.
- **Morale:** the highest levels of a department's morale gave too much extra effort. They give less now, and the lower ones are unchanged.
- **Sponsors:** in the commercial sponsors page, the contracts of more than one year with a supplier were shown as works deals when they were not. They now show the level that was signed. As far as I can tell it was a display problem and not a data one.

The 1998 data

Everything so far has been about behaviour. This chapter is about the starting point: who drives, who runs the teams, what the circuits are like, and how good the engines are on day one. The goal was to have a grid that feels like the real 1998 season, without breaking the balance the original game was built on.

Most of the data is untouched, and it stays as it is in 1.03. What follows is only what has moved.

The drivers

The starting skills of the race drivers are the 1.03 ones, with a short list of exceptions. Each of those is a deliberate call:

- Some drivers go up: Fisichella, Barrichello, Trulli, Salo, Takagi, Nakano and Tuero are all a bit better than they were. Villeneuve and Trulli got an extra adjustment on top of that.
- Some go down or change shape: Irvine is a little less sharp, Hill's profile is reshaped, and Panis and Diniz are rebalanced.

The test drivers are a bit better in general than in the original, which was too stingy with them: Montoya, Badoer, Zonta, Sarrazin, Collard and Montermini. The test driver seats of Benetton, Jordan and Stewart, added with Veedub's GPWEdit, remain.

None of this is a statement about who was better in real life. It's a starting grid for a game, and if you disagree with a rating, I'd like to hear why. There is also an extra group of free drivers, the ones who are not in Formula One at the start: Wilson, Heidfeld, De la Rosa, Watson, Verstappen, Zanardi, Gene, Junqueira, Rodriguez, Watt and Ayari. They were rewritten with their 1998 wages, bonuses, ages, nationalities and skills. Two of them come with money behind them, as pay drivers.

The people in the pit wall

The department chiefs - Commercial Manager, Chief Designer, Technical Director and Chief Mechanic - are paid better than in 1.03, nearly all of them, with bonuses to match. In some cases their level and their age have changed too. The effect is that a good chief costs something real, so hiring one is a decision with consequences for the budget. And in the case of the CD and TD, their skills are not so much based on my opinion about them than in trying to "preview" the 1999 season: there's some random introduced since Veedub's 1.03 patch changed the formula, but you should expect a pretty strong season for Jordan and Stewart in 1999, for instance.

The circuits

The circuit data is the part of the 1998 data that you'll notice without looking for it. The overtaking value has been made tougher at almost every circuit, particularly so at Imola. The default rain setting of the set-up has also moved on most of them, and Japan most of all. And two circuits have slightly different race distances: France and Japan, but that one is also coming from GPWEdit's fixes.

The engines

In the original, the eight engines were balanced so that every one of them added up to the same total: if an engine was strong in one place, it was weak in another. That made them hard to tell apart.

Now the engines are not equal. Some have a higher total than others, and the gap is real: Mercedes Benz and Ferrari are at the top, and Hart is clearly the weakest. If that sounds unfair, well, it's 1998. You can still compensate with the rest of the car, but you start from a different place depending on who supplies you.

The starting chassis (new in 0.2)

I have slightly adjusted the starting chassis handling of five teams: Ferrari, Benetton, Jordan, Stewart and Minardi, to my own handling table. It is the same idea as with the drivers: a starting grid that feels like 1998, where the order of the cars at the start of the first season is the one you would expect.

The performance curve

The curve that converts the car's chassis and engine into raw performance is smoother than in the original. The old one had a steep stretch in the middle, where a single point of handling bought you a huge jump, and was nearly flat at the top. The new one tapers gradually, so there is always something to gain from developing the car, wherever you are on the scale. The lowest and the highest ends remain exactly where they were.

The headquarters

The headquarters now hold fewer people at every level and cost more to run. It stays for now, but it's one of the first things I'm prepared to revisit once people have played with it, so if it feels like a punishment, say so.

Sponsors and suppliers

The tobacco brands are the 1998 ones: Benson and Hedges, Gauloises, Marlboro, Mild Seven, West and Winfield, with the right names in the text. The list of sponsors and who starts with whom is the same as in the original; what changes is how big a name some of them are.

Two numbers describe each sponsor. The rating is how important it is in the game's eyes, and with it how much it pays and how hard it is to win over. For suppliers there is also a class, which is how fast they develop their product year after year. Among the team sponsors, Benson and Hedges and Red Bull now rate higher than before, and Gauloises and Winfield a bit lower. Mugen-Honda and Peugeot rate lower as engine suppliers. As for development, Bridgestone now develops faster than Goodyear, which was the other way round, and Ford and Peugeot develop their engines more slowly than before. That matters more than it sounds: when the AI chooses an engine or a fuel supplier it goes for the ones that develop fastest, so these changes can shift who ends up with what over a few seasons.

News and texts

The news in GPW is one of the things that gives the game its character, and also one of the things that wears out first. After a couple of seasons you've read every headline several times. This chapter is about making the news a little more alive, and fixing a handful of texts that were simply wrong.

The same event, told differently

In the original, every event has a single wording. A driver leaves a team, and the headline and the story are always the same. I've added a family of different versions for each of them, written in the same style as the originals with a lot of help from *ClaudIA* , my pet name for Claude Code. The game now picks one text at random whenever it needs to tell the story, together with the headline that goes with that version. When a text hasn't been translated into your language, the game falls back to the original, so nothing is ever blank.

This doesn't change when a news item appears, only how it reads. The English texts are done. French and German are not up to date, so they are not in the package and those languages keep the original game's texts: I'd rather leave them alone than put in something I can't check.

New events with their own texts

The things that didn't exist before now have their own wording: the Safety Car, the investigation and the penalty, the two-hour limit, the mid-season termination of a contract. They are written to sound like the original game stuff, and the Safety Car stories in particular come in a few shapes depending on what the strategy of the team was.

New in 0.2: the news of a signing as a test driver now says whether the driver has been signed as a test driver or as a race driver, with wordings and headlines of its own.

Winners, margins and the "narrowly" headline

The news about the race winner used to distinguish between a comfortable win and a close one. At least, that was the intention. The code compared the gaps the wrong way round, and the check always came out the same, so the "crushed", "dominated" and "narrowly" headlines could never be reached. Now they can. And a close win is reported as such for every team: before, only the AI got the "narrowly" version, while the player's team was told it had simply won again.

Rumours about the chiefs

The rumour that a department chief might change teams had its job titles shifted: the Chief Designer got the story meant for the Technical Director, the Technical Director got the Chief Mechanic's, and so on. On top of that, the names of the person and the team came out the wrong way round for the Chief Designer. And when the affected team was yours, the check for one of the jobs was wrong and the Chief Mechanic never made the news at all. All three are fixed.

Injuries

The news and the mail about an injury now tell you how many races the driver is going to miss. It's a small thing that I always wanted when a driver went to hospital. It needs the updated text files to show, and without them the game works as it did.

Lost sponsors

When you lose a deal with a sponsor that pays you in money, the game used to send the mail meant for a supplier ("Lost Works Deal") or the one for a partner. Now you get the one that fits, the one the game already had for sponsors.

The names of 1998

The names of the tobacco sponsors are the real 1998 ones, and so are several drivers and team names that appear in the text. One of the headlines I changed is the one that used to say "GAME OVER", which now says something a bit friendlier. I'm not going to tell you what, because it's much better when you stumble across it.

Screens, speed and the map

This chapter collects the changes you see rather than feel: the speed of the simulation, the panels and the map. None of it changes the result of a race, but a good deal of it changes how pleasant it is to follow one.

Watching a race faster

The speed selector in the original ran from slower than real time to a bit faster. For a game where a race can be a long afternoon, it didn't help much. The selector now goes well beyond the old maximum, and it moves in much bigger steps with each press of F8 and F9, so you can get to the speed you want in a couple of presses.

The way it works has changed as well. The game runs several simulation steps for each frame it draws, rather than stretching time, and a fraction is carried between frames so the speed is smooth. The selector starts at the value I choose when I build the EXE, and the label on the screen matches what you get.

The race order, with gaps

The "Race Order" panel used to list only the cars that had crossed the line on the current lap, together with their laps. Now it always lists the whole field, in the live order of the race, over two pages. It has four new columns: the stops each car has made, the tyre it is on now (D for dry, I for intermediate, W for wet, the last two in green and red; soft and hard are not told apart), the gap to the leader and the gap to the car ahead. Names are surnames, with an initial only when two drivers share one - so you see "M. Schumacher" and "R. Schumacher", and "Frentzen" instead of "H.Frentzen".

Pit stops, lap by lap

The "Pit Stops" panel used to give only the number of stops and the fastest and slowest one. Now it lists every driver who has stopped, in the live race order, with three blocks of columns: Pit 1, Pit 2 and Pit 3. Each block has the lap of the stop, with an L in front of the number (repetitive, I know, but clearer at a glance), and its duration in seconds, rounded to the tenth ("27.6s"). The game never kept those two numbers, so the pit stop routine now writes them down for each car's first three stops. Names follow the same rule as in the Race Order.

Lap times of any driver

The "Lap Times" panel only ever showed your own cars. Now you can move on to any driver on the grid with the page arrows on the screen: at the end of one driver's pages, the next press takes you to the next driver in the race order, and the other way round. The page keys on the keyboard do the same job where they reach the game.

Circles that move like cars

On the race map each car is a circle. The circles are only a drawing, but they had a habit that bothered me: on some stretches, usually the long straights after a corner, they would cross in an instant, and on others they crawled. The map had been drawn without looking at how long the cars really take on each section of the track.

I redid it circuit by circuit. Each circle now moves along the map at a pace that follows more closely how long the cars take on every stretch, and the jumps at the boundaries between sections are gone or much smaller. The straights are still faster than the corners, as they should be. It’s not perfect, but it makes a noticeable difference (or at least, I notice it). The cars themselves, their speed and their lap times, don't change at all: this is only the drawing.

Qualifying times

The "Qualifying Times" panel gets a column for the laps each driver has done, shows surnames in the same way as the race screen, and the columns have been rearranged to fit.

Which windows open, and where

At the start of a session the game opens a standard set of windows. The old default included the TV feed and the camera; the new one opens the map, the classification panel, the circuit conditions and the commentary transcript, clockwise from the top left. And the game now remembers how you left your four windows at the end of a session and puts them back at the start of the next one of the same kind. The first time, you get the default layout.

Car Internals (new in 0.2)

For each of your drivers, a bar for each of the seven large parts of the car (brakes, clutch, electronics, gearbox, hydraulics, suspension and throttle), showing how much is left of each.

Live Championship Standings (new in 0.2)

A new panel in the race screen, reached with the arrows of the panel after Car Internals. It has two columns: the drivers' championship and the constructors'. They show the table as it would be if the race ended at that moment: the points of the season plus the points of the position each driver is holding now, without counting the cars that have retired. Each row has its place, the name and the points. Your cars and your team are in yellow.

Next to the place there is an arrow: green and up if the driver has gained places compared with the championship before the race, red and down if he has lost them. It only appears if somebody has really been passed, so two drivers with no points don't move even if others score.

The Position tab (new in 0.2)

It used to show two cars ahead and two behind. Now it shows five on each side, with the window sliding so that the panel is always full.

Race Order at the start (new in 0.2)

At lap zero the panel showed an order that looked random. It shows the grid now.

Saved games (new in 0.2)

If you saved a game in the middle of a race, the times of the pit stops came back as zero when you loaded it. The saved game now carries them, together with the other data that the new parts of the game use during a race. Games saved with 0.1 still load, with those figures at zero. An SC that is on track at the moment of saving is not kept.

The map

The maps of the circuits have a full pit lane. The cars stay visible on it, and the Safety Car has its own silver marker. The details are in the pit lane chapter.

Small fixes

These are the things that don't deserve a chapter, but that I'd rather list than hide.

Loading a saved game. When you loaded a game, the game forgot how many set-up points your team had already used. Two things followed. The buttons to lower points were disabled, and the weekly process decided you hadn't used any, sent you a stern mail about unused testing data, and took some morale away from your chiefs and drivers. It now rebuilds that count when the game loads, together with the testing time of each driver, so a loaded game behaves like the one you saved.

A leak in the sound. The sound code was opening system events for every sound it played and never closing them. In a long race the number grew into the thousands. I've fixed the two places I found, and it's a lot better. I'll be honest: the growth hasn't stopped completely. There is a second source I haven't tracked down, so for now it's much slower and not zero.

A broken wing, a lost stop, a shuffled lane. These are covered in the chapters on the AI and the pit lane, but they belong in the same list of things that were plainly wrong: the AI ignoring a broken front wing, a postponed stop that vanished, and cars leaving the pit lane in the wrong order.

Rumours, margins and mails. The news that came out with the wrong text, the headlines nobody could ever read, and the mail you shouldn't have received are in the news chapter.

The constructors' table. Teams on no points all took the same stored position, which fed the share prices and the news with the wrong values. They each get their own now.

Takagi and Rosset. (New in 0.2) In the game, Takagi was called Rosset and the other way round. I swapped their voice files, so each driver is now called by his correct name.

Plenty of the changes in this mod are fixes in disguise. I try to list them all, because I think you should know when something that looked like a quirk of the game was in fact a mistake.

What comes next

Things on my list

Some of what is left is small, some is not:

- The leak in the sound is smaller but not closed: there is a second source I haven't tracked down.
- The times the AI uses to decide whether a stop is free under the SC are estimates, and a few circuits still use a general value for the pit lane.
- I'm not convinced the minimum stint length should also apply to the last stint of the race. I may relax it, or let that one go.
- The wind is a first version and I'll take another look at it once more of you have played with it.
- The overall pace at Germany still needs a look, now that the spread is right.
- There is a report that the ghost car sometimes misses the pit entry and stays out on track. What I changed attacks the jump, not what happens afterwards.
- The French and German versions of the new texts are missing, and the package does not include those two languages.
- Some of the 1998 numbers are my best reading of the real season and not a measurement. If you think yours would be better, please tell me.

Ideas

I have more ideas than time, which is a nice problem to have. A few that come back to me all the time: rotating the AI's tyre sets in qualifying, accident rates that depend on the circuit, a proper rework of dry and wet conditions, a choice of suppliers, more variety in the pool of drivers, a calendar with more races, and the community's own proposals. People have already sent me some good ones, and there's a page where you can see them and add your own: GPW Evolution '98 · Send a proposal | Enviar una propuesta

Reporting bugs

Overall I want the same thing I wanted at the start: to keep the GPW experience and fix what was broken. If something in here doesn't feel right to you, or if you find a bug I haven't noticed, I'd like to hear about it, with the circuit, the round and what you were doing: on the forums at GPW General | GPM Planet, at the forum's Discord, or wherever you find me. It's the only way this gets better.

Happy gaming!